Introducing the Module

CS5701 - Quantitative Data Analysis

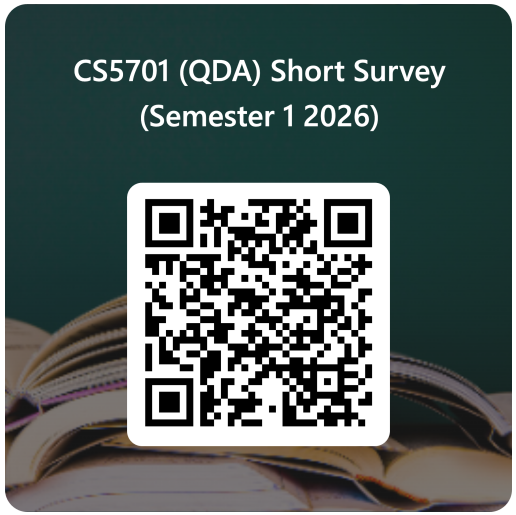
Dr Isabel Sassoon

Department of Computer Science
Brunel University London
isabel.sassoon AT brunel.ac.uk

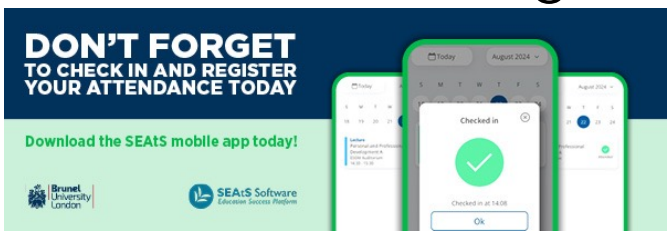
30th September 2026

Welcome to QDA – While we wait

If you have not already... can you fill this in?



.... and check in using the **SEAtS App**



Welcome to QDA – House keeping rules

- I will start at 11:05 prompt
- The Lectures are being **recorded** - the recordings will be shared but it can take a few hours to make them available
- There will be time for questions during the sessions
- All the material for this module - CS5701 is and will be shared on BrightSpace in the dedicated module page.



Welcome to QDA – The Team

- Isabel Sassoon – (Module Leader)
- Sarath Dantu and Leila Ghanbar
- A GTA team will also support you in the labs

Welcome to QDA – Isabel Sassoon – About me

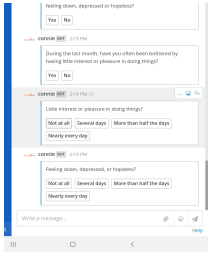
- "Doing cool and helpful things with data"
- Statistician - Fellow of the Royal Statistical Society
- Data Science and AI Research interests:
  - Explainable data-driven models
  - Computational Argumentation and Large Language Models
  - Health Informatics
  - Model lifecycle and drift
- See my website for more details: https://www.brunel.ac.uk/people/isabel-sassoon

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Welcome to QDA – Isabel Sassoon – About me

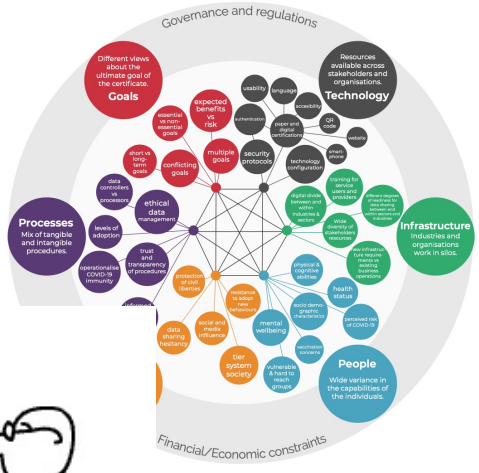


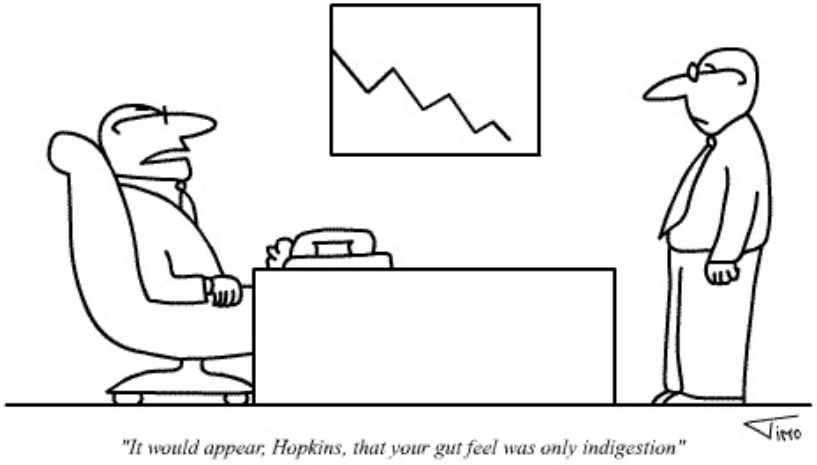












QDA - The weekly schedule

Each week you will have:

- A one-hour Lecture on Wednesday 11 – 12 in **MCST 054**
- A two-hour lab session on Wednesday 3 - 5 pm in **HWLL 302**.
- The lab is in 2 parts:
  - Part 1: A guided lab session
  - Part 2: Time for independent practice with a chance to ask us questions

Note: Week 6 is reading week.

QDA The weekly schedule (continued)

- During Part 2 of the Labs you will be provided with worksheets and data, in order to independently practice and apply the weeks' concepts
- These are available to download from CS5701 Brightspace
- We will be available to answer questions and support you
- We will share solutions to these the week following their release
- There will be scheduled on-line drop in sessions if you have questions on any of the material or the module or just to say hello!
- On there is a discussion forum on Brightspace and you can always email me: isabel.sassoon@brunel.ac.uk

QDA The weekly schedule (continued)

For each week there will be this information shared on the Brightspace module page:

- Suggested pre-reading material - for you to familiarise yourself before the lecture
- The Lecture slides
- Additional material to provide more depth in certain areas - this is available for those wanting more information
- The code and data for the lecture and labs (if applicable)
- The independent worksheet for you to work through in the second part of the lab
- Independent worksheet solutions from the previous week

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QDA Learning Outcomes

- LO1: Develop understanding of statistical data analysis methods and their relevant use in data science application and research.
- LO2: Learn to critically evaluate the strengths and the weaknesses of statistical data analysis methods alongside an understanding of how and when to use or combine methods.

For more details on the module head to the Module Page on Brightspace where you will find a copy of the study guide.

QDA - Module Overview

"The numbers have no way of speaking for themselves. We speak or them. We imbue them with meaning."
– Nate Silver, The signal and the noise

In the module we will step through the following stages:

- Exploring one variable at a time
- Asking questions regarding one variable
- Exploring and modelling the relationship between variables

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QDA Week by week

The lectures and labs have a weekly topic:

Week	Topic
1	Exploring Data
2	Inferential Statistics – Part 1
3	Inferential Statistics – Part 2
4	Count Data and Table Analysis
5	Linear Regression
6	READING WEEK
7	Analysis of Variance (ANOVA)
8	Multiple Regression
9	Generalised Linear Models (GLM) and Logistic Regression
10	QDA in the wild
11	Revision Lecture

Table: Lecture Schedule

QDA Labs

- We will be using R in Rstudio
- R is widely used for Statistics and Data Science
- R Studio and R are both open source and are available for Windows, Mac and Linux
- You will have access to R and R studio on the lab PCs

Expectations on AI use

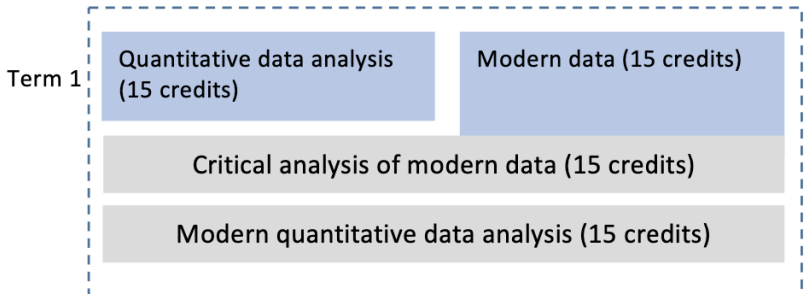
- The University has general guidance on using artificial intelligence in your studies.
- Generative AI and Large language models (LLMs) all have the potential to answer questions on code and material relevant to this module.
- Sometimes the answers are of very high standard with helpful accompanying explanations, sometimes they are wrong.
- A difficulty for a learner is to know the difference.
- **If you do not learn how to do quantitative data analysis yourself then it will be difficult for you to know the difference between a correct or incorrect answer.**

Advice

- **QDA is not a spectator sport**
- Mistakes are part of learning (practice, practice ...worksheets are not marked or assessed)
- Using cheat sheets, asking for help, and searching online are not hacks just for beginners but something that even professional R users do
- When you use resources - reference and acknowledge them (including GenAI)

Assessment blocks vs teaching blocks

CS5701 is a teaching block which has **integrated assessment blocks with CS5702 (Modern Data)**



You will have two assessments:

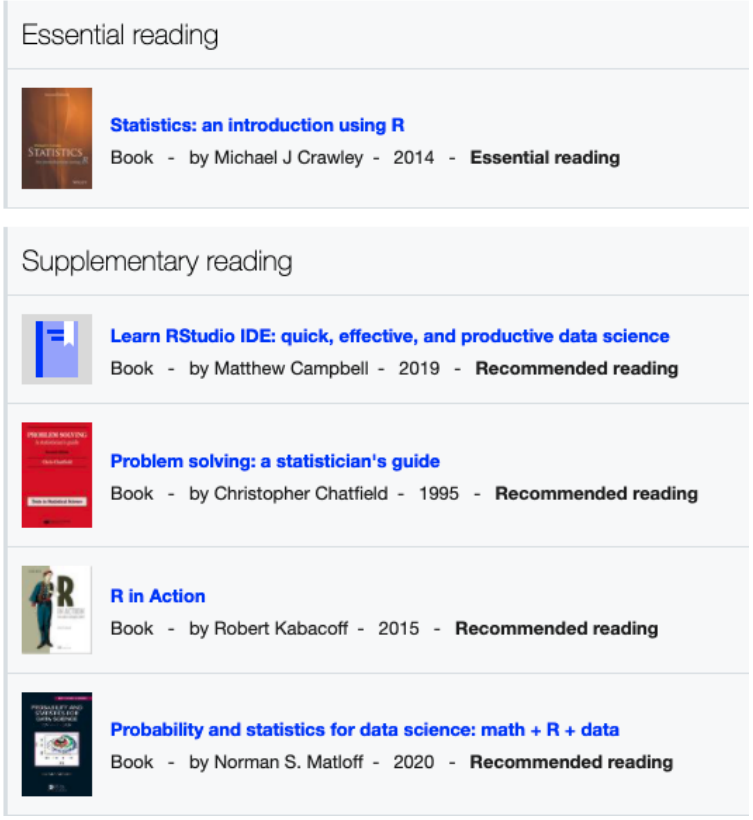
- coursework (CS5801)
- an exam (CS5802)

These will cover the material in CS5701 (this module) and CS702 (Modern Data).

QDA Reading list

The main text book is:

- Crawley, M.J., 2015. Statistics: an introduction using RJ Wiley, England.
- It is available (as all others) as digital downloads through the library



Questions?

- We try to have time for questions but if we don't get to your question in the lecture
- Post it on the Brightspace discussion board
- ... or ask them during the second part of the lab session later today.
- We will answer them!!

We will now move to week 1 material for QDA ....